Table 7.8. Differences between normal cell and cancer cell

Normal Cells		Cancer cells	
Small, uniformly shaped nuclei Relatively large cytoplasmic volume			Large, variable shaped nuclei Relatively small cytoplasmic volume
Conformity in cell size and shape Cells arranged into discrete tissues			Variation in cell size and shape Disorganised arrangement of cells
May possess differentiated cell structures Normal presentation of cell surface markers			Loss of normal specialised features Elevated expression of certain cell markers
Lower levels of dividing cells Cell tissues clearly demarcated			Large number of dividing cells Poorly defined tumour boundaries

not invade other body parts. In the malignant tumour, the cells grow indefinitely, detach and migrate into healthy surrounding tissues.

In normal cells, cell growth and differentiation is highly controlled and regulated. But in cancer cells, there is breakdown of this regulatory mechanism. Normal cells show a property called **contact inhibition**, which inhibits uncontrolled growth. Cancer cells do not have this property. As a result, cancerous cells divide continuously giving rise to mass of tissues called tumours **(Table 7.8).**

When a cell undergoes malignant transformation, it acquires new surface antigen and may also lose some normal antigens. These antigens are present on the membranes of malignant cells and they induce an immune response. Both humoral and cellular responses can be observed in malignancy. Cancer cells can avoid immune detection as they are not foreign bodies but are abnormally functioning body cells. This makes them difficult to treat.

The concept of immunological surveillance postulates that the primary function of the immune system is to "**seek and destroy**" malignant cells that arise by somatic mutation. The efficiency of the surveillance mechanism reduces either as a result of ageing or due to congenital or acquired immunodeficiencies, leads to increased incidence of cancer. Thus, if immunological surveillance is effective, cancer should not occur. The development of tumour represents a lapse in surveillance.

Scope of Immunology

The younger graduates in this field can find number of employment opportunities in Government as well as private hospitals. The scope of the immunology is immunotherapy, microbial immunology, clinical immunology, cellular immunology, allergy and immunology, translational immunology, transplantation immunology, neuro-inflammatory disorders, tumour immunology, vaccine immunology, inflammatory disorders, ocular immunology and inflammation.

Immunotherapy of cancer

Immunotherapy also called biological therapy uses substances made by the body or in a laboratory (monoclonal antibodies) to improve or to resist the immune system function. Different approaches have been attempted in the immunotherapy of cancer. Immunotherapy appears to be important in

XII Std Biology-Zoology Chapter-7 EM.indd 129 23/12/2022 12:40:00